

The Algorithmic Nexus: An Econometric and Historical Analysis of Artificial Intelligence and Labor Demand

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Abstract

This report provides an analysis of the impact of Artificial Intelligence (AI) on labor demand, synthesizing theoretical frameworks from labor economics thanks to recent studies, historical perspectives on information networks, and cutting-edge empirical evidence from France. We investigate if AI represents a continuation of skill-biased technological or a fundamental rupture leading to the obsolescence of the "human". Drawing on the "Task-Based" model of Acemoglu and Restrepo, we dissect the competing forces of displacement and productivity effects, according to the derivation of the conditions under which automation creates a "useless class" versus those under which it reinstates labor into new higher-value tasks. We integrate the philosophical insights of Yuval Noah Harari regarding his famous book *Nexus* and the role of bureaucracy and information networks, juxtaposing the historical function of human administrators with the emerging capabilities of algorithmic agents. The core of this analysis focus on a detailed examination of the seminal work by Aghion et al. (2025), utilizing comprehensive French firm-level data to estimate the causal impact of AI adoption on employment, sales, and skill composition.

1 Introduction

1.1 The Renaissance of the Machinery Question

The "Machinery Question" is a crucial point that has haunted economic/economists since the Industrial Revolution. While 20th-century economists largely dismissed these fears, believing that technological progress simply shifted labor toward more productive roles, the emergence of Artificial Intelligence has shattered that consensus and creates an inevitable earthquake.

Unlike previous industrial revolutions that automated physical "muscle power," AI represents a fundamental rupture because it targets non-routine, cognitive, and creative tasks that were once considered the exclusive domain of human intelligence. This shift challenges the very definition of "skilled" labor. The "Horse Equilibrium" drawing first by Leontief tell: a specific economic state where a workforce is rendered obsolete because technology has advanced to the point where the cost of maintaining that workforce exceeds its economic output. Ultimately, we may be facing a structural phase shift leading to the rise of a "useless class": a population rendered unemployable not by a lack of demand but by a lack of economic utility compared to AI and so less rapid.

1.2 Defining the Scope: From Bureaucracy to Algorithms

To understand the economic impact of Artificial Intelligence, the report defines it broadly as an information-processing technology that includes machine learning and natural language processing. By applying Yuval Noah Harari's thought from *Nexus*, the analysis suggests that human dominance has always relied on information networks managed by bureaucracies. Historically, these "biological computers" (clerks and administrators) were essential for maintaining the "nexus" of shared fictions, such as laws and money, that hold society together.

However, AI represents a fundamental rupture in this historical structure because it is not just a tool, but an agent capable of making decisions and generating ideas replacing humans. In economic terms, AI is now automating the specific "bureaucratic" functions that protect the white-collar middle class. This transition from human-centric networks to inorganic intelligence forces a rigorous re-evaluation of how we calculate labor demand in the modern era.

1.3 The French Context: Rigidities and Innovation

This part and the report focus specifically on evidence from France. France represents a highly regulated labor market with significant structural rigidities, distinct from the flexible labor markets of the US or UK usually studied in this context. France is characterized by:

- **Strong Employment Protection:** High firing costs and strict labor contracts.
- **Size-Contingent Regulations:** A proliferation of labor laws that trigger specifically when firms cross size thresholds (most notably at 50 employees). This creates an "Innovation Valley," where firms suppress growth to avoid regulatory burdens.
- **Centralized Data:** The availability of high-quality, matched employer-employee administrative data (via CASD) allows for a level of granular analysis impossible in many other jurisdictions.

2 Theoretical Framework: The Task-Based Model of Automation

To move beyond anecdotal evidence and alarmist headlines, the analysis is grounded in a formal economic model designed for precise predictions. The author argues that standard 20th-century models, such as the Cobb-Douglas production function where basically $Y = AK^\alpha L^{1-\alpha}$, are insufficient for analyzing AI. In these traditional structures, technological progress is typically modeled as an increase in A (Total Factor Productivity) or labor-augmenting efficiency, which imposes a mechanical complementarity where better technology always increases the marginal product of labor. This essentially ignores the core phenomenon of labor being replaced in specific tasks.

Instead, the study utilizes the Task-Based Framework developed by Acemoglu and Restrepo in 2018. This model treats the production process not as a combination of factors, but as a continuum of specific tasks. By adopting this lens, it becomes possible to move away from the assumption that technology is always beneficial to labor and instead analyze the specific conditions under which human labor is substituted by algorithmic agents.

2.1 The Production Function of Tasks

Consider an economy where a final good Y is produced by combining a continuum of tasks, indexed by $i \in [N-1, N]$. The production function is given by a Constant Elasticity of Substitution (CES) aggregator:

$$Y = \left(\int_{N-1}^N y(i)^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}} \quad (1)$$

Where:

- $y(i)$ is the output of task i .
- σ is the elasticity of substitution between tasks.
- N represents the range of new tasks created by innovation.
- The interval $[N-1, N]$ represents the set of tasks currently required for production.

Tasks can be performed by either labor (L) or capital/AI (K). The defining feature of the model is the threshold of automation, denoted by I . Tasks $i \leq I$ are technically automated and can be performed by capital. Tasks $i > I$ are complex and must be performed by labor.

The production function for a specific task i is:

$$y(i) = A_L \gamma_L(i) l(i) + A_K \gamma_K(i) k(i) \quad (2)$$

Where A_L, A_K are factor-augmenting productivities, and $\gamma_L(i), \gamma_K(i)$ are comparative productivity schedules. We assume $\gamma_L(i)/\gamma_K(i)$ is increasing in i , meaning labor has a comparative advantage in higher-index (more complex) tasks.

2.2 The Automation Decision

The crucial economic decision for a firm is not just technical feasibility, but economic viability. A task $i \leq I$ will be automated if and only if the unit cost of using capital is lower than the unit cost of using labor. This yields the automation condition:

$$\frac{W}{A_L \gamma_L(i)} > \frac{R}{A_K \gamma_K(i)} \quad (3)$$

Where W is the wage rate and R is the rental rate of capital. This inequality highlights that automation is endogenous to wages. If wages (W) are high, the incentive to automate increases. Conversely, if labor is cheap (or suppressed), firms may delay automation even if the technology exists. This dynamic helps explain the “Innovation Valley” in France.

2.3 The Displacement Effect

When AI advances, it extends the threshold of automation from I to I' . This creates a Displacement Effect. Capital takes over tasks previously performed by labor. Mathematically, the change in labor demand depends on how this shift affects the labor share (s_L). The labor share in this model is derived like:

$$s_L = \frac{WL}{Y} = (1 - \eta)(N - I) \left(\frac{W}{A_L \gamma_L} \right)^{1-\sigma} \quad (4)$$

Here, an increase in I (automation) basically and directly reduces the term $(N - I)$, which represents the range of tasks allocated to humans.

- **Direct Impact:** Automation ($dI > 0$) always reduces the labor share of value added.
- **Wage Impact:** It reduces the demand for labor. If this displacement effect is strong enough, it can reduce real wages (W/P) even if total productivity rises.

2.4 Countervailing Forces: Productivity and Reinstatement

However, the Acemoglu-Restrepo framework identifies two critical countervailing forces which historically have prevented the “end of work.”

2.4.1 The Productivity Effect

By switching to cheaper capital thanks to AI, the firm lowers its unit costs of production. In a competitive market, this lowers prices (P) of course and expands demand for the final good Y :

$$Cost_{reduction} \rightarrow Price \downarrow \rightarrow Demand_Y \uparrow$$

If the demand for the good is elastic, the firm produces more. Crucially, to produce more Y , the firm needs to perform all tasks in the production chain, including the non-automated tasks ($i > I$).

- **Result:** The expansion of scale increases the demand for labor in the remaining human tasks.

- **Empirical Sign:** This is the primary mechanism tested by Aghion et al. (2025). If they observe that AI adoption leads to sales expansion and employment growth, it confirms that the *Productivity Effect dominates the Displacement Effect*.

2.4.2 The Reinstatement Effect (New Tasks)

Technology does not just automate; it creates. The expansion of the task frontier from N to N' creates entirely new tasks in which labor has a comparative advantage (newly “Prompt Engineers,” “AI Ethics Compliance Officers”, ...).

$$\frac{ds_L}{dN} > 0 \quad (5)$$

The creation of new tasks creates a powerful “Reinstatement Effect” that raises the labor share and increases the demand for labor, finally counteracting automation.

2.5 The “Horse Equilibrium” vs. Human Resilience

A critical theoretical insight involves the “Leontief analogy” regarding horses. In the early 20th century, the internal combustion engine displaced horses from transport and agriculture. Horses experienced a massive displacement effect but zero reinstatement. They reached a “Horse Equilibrium” where their equilibrium wage (cost of fodder) was higher than their marginal product, leading to a collapse in the horse population.

The existential question for AI is with two different views: **Are humans like horses?!**

- **Optimistic View (Aghion/Acemoglu):** Humans have a comparative advantage in “general intelligence” and creativity. As AI automates routine tasks, humans move to N' (new complex tasks).
- **Pessimistic View (Harari):** AI is not just a muscle; it is a mind. If AI eventually dominates humans in cognitive tasks (shifting I all the way to N), there is no “safe ground” for reinstatement. The “Horse Equilibrium” becomes the “Useless Class” equilibrium.

3 Historical and Philosophical Context: The Bureaucratic Nexus

To interpret the empirical data we will examine later, specifically the negative impact of AI on “administrative” employment and so must contextualize the role of administration in human history. We turn to the work of Yuval Noah Harari and the historical theories of money and bureaucracy.

3.1 Information Networks as Power Structures

In *Nexus*, Harari argues that the defining trait of *Homo Sapiens* is not individual intelligence but in fact the ability to build massive networks of cooperation. These networks are held together by information.

- **The Fiction:** Cooperation requires shared stories (gods, nations, money).
- **The Bureaucracy:** Maintaining these fictions requires processing immense amounts of data.

Historically, the “technology” used to process this data was the human bureaucrat. The invention of writing in Mesopotamia was primarily for accounting—tracking taxes, debts, and grain. The “clerk” is a biological component of an information processing machine.

3.2 Trusted Third Parties and the Cost of Trust

This historical view aligns with the economic theory of money and intermediaries.

- **Money as a Ledger:** Money emerged not just to facilitate barter, but as a fiscal technology to track debt and surplus extraction by central authorities.
- **Trusted Third Parties (TTP):** Economic systems rely on TTPs (banks, notaries, state administrators) to verify identities and enforce contracts. These intermediaries are essentially “bureaucrats of trust.”

Labor regulations and corporate structures created a massive demand for these intermediaries. In France, the complexity of labor law (the *Code du Travail*) necessitates armies of HR specialists, payroll administrators, and compliance officers. In the Acemoglu framework, these are tasks in the $(N - I)$ range.

3.3 AI as the Ultimate Bureaucrat

The transition to AI represents the automation of this bureaucratic function. AI is, fundamentally, an information processing technology. It excels at the tasks that define bureaucracy: sorting data, recognizing patterns in lists, and applying rules to specific cases (sometimes complex for humans or too big).

Harari distinct this transition period from the industrial revolution. We are not automating “muscle”; we are automating “order.”

- **The Threat:** If AI takes over the maintenance of the social nexus (deciding who gets a loan, who gets hired, who gets audited), we replace human judgment with “alien intelligence.”
- **The Economic Implication:** The “Administrative Process” tasks are the low-hanging fruit for AI.

Studies predict a sharp decline in demand for the white-collar middle class—the clerks, administrators, and mid-level managers who historically managed the information network.

3.4 The “Useless Class” Paradox

The most dystopian implication of this transition is the creation of a “useless class.” If the economic value of human labor in both physical (robotics) and bureaucratic (AI) tasks drops below the cost of subsistence (or the minimum wage), we harvest social contract fractures.

Harari notes that unlike the 20th-century working class, which was his contribution was important for military and economic power, the “useless class” of the 21st century might be irrelevant to the elites. The interaction with the Acemoglu model is clear: if I (automation) advances faster than N , and if the “Innovation Valley” regulations prevent firms from hiring humans for new marginal tasks, the “useless class” grows.

4 Data and Methodology: The French Laboratory

To test these theoretical fears against reality, we turn to the comprehensive study by Aghion, Bunel, Jaravel, Mikaelson, Roulet, and Søgaaard (2025). This paper provides a rigorous empirical test of the AI impact in France.

4.1 Data Architecture

The robustness of the study stems from its linkage of three high-quality administrative datasets, accessed via the **CASD (Centre d'accès sécurisé aux données)**, a highly secure environment for French confidential data.

4.1.1 The Firm-Level Financial Data (BIC-IS)

- **Source:** *Bénéfices Industriels et Commerciaux* (DGFIP), **Content:** Comprehensive tax records for all firms subject to the normal tax regime, **Key Variables:** Value Added (VA), Sales (Turnover), Capital Assets, Intermediate Consumption, **Role:** Allows the calculation of firm-level productivity (TFP), growth, and size.

4.1.2 The Matched Employer-Employee Data (BTS - DADS/DSN)

- **Source:** *Base Tous Salariés* (Insee), **Content:** A universe file of all employee contracts in the private sector, **Granularity:** Data is at the establishment level (SIRET), tracking exactly who is being hired or fired.
- **Key Variables:**
 - *Occupations:* Detailed occupation codes (PCS-ESE 4-digit).
 - *Skill Levels:* Decomposes the workforce into Managers (*Cadres*), Intermediate Professions, Clerks, and Blue-Collar workers.
 - *Wages & Hours:* Detailed hourly wage data for labor share calculations.

4.1.3 The ICT Survey (Enquête TIC)

The *Survey on Information and Communication Technologies* (Insee / Eurostat) is the primary source of the “treatment” variable. Starting in 2018 which refers to 2017, survey explicitly asks firms if they use Artificial Intelligence, breaking down usage into 7 specific categories: Marketing or Sales, Production Processes, Administration Processes, Management of Enterprises, Logistics, ICT Security, HR Management or Recruiting

4.2 Measuring AI Exposure: The Bergeaud (2024) Index

Beyond the binary adoption variable, the authors use a continuous measure of potential exposure constructed by Bergeaud (2024). The methodology involves feeding task descriptions from the O*NET database into GPT-4 and just this AI to evaluate their “substitutability” by Generative AI.

These task scores are aggregated to the occupation level using task importance weights $p(t)$:

$$s1(p, P) = \frac{\sum_{t \in T} s1(t, P) \cdot p(t)}{\sum_{t' \in T} p(t')} \quad (6)$$

This continuous variable allows the researchers to test if firms with more exposed workforces adopt AI more aggressively or not.

4.3 Econometric Strategy

The authors face a classic endogeneity problem: do firms grow because they adopt AI or adopt AI because they are growing? To solve this, they employ a rigorous identification strategy.

4.3.1 The Dynamic Event Study

To check for “parallel trends” and so they estimate the following dynamic equation:

$$y_{it} = \sum_{k=-K}^L \beta_k D_{it}^k + \mu_i + \delta_{st} + \epsilon_{it} \quad (7)$$

Where:

- y_{it} : Outcome variable (Log Employment, Log Sales).
- D_{it}^k : Dummy variables relative to the adoption year ($k = 0$).
- μ_i : Firm fixed effects (absorbing time-invariant characteristics).
- δ_{st} : Industry-by-Year fixed effects (absorbing sector shocks).

Validation requires that pre-adoption coefficients (β_{-3}, β_{-2}) are statistically indistinguishable from zero only.

4.3.2 The Difference-in-Differences (DiD) Specification

To summarize the average treatment effect, a DiD model is used:

$$y_{it} = \beta(Adopt_i \times Post_{it}) + \mu_i + \delta_{st} + \gamma X_{it} + \epsilon_{it} \quad (8)$$

Where β measures the causal semi-elasticity of AI adoption on the outcome.

4.3.3 The Shift-Share Instrument (SSIV)

As a robustness check against for omitted variable bias, the authors construct a Shift-Share Instrument Variable. A firm is more likely to adopt AI if its pre-existing workforce performs tasks that are becoming cheaper to automate than human workforce:

$$Z_{it} = \sum_o \frac{L_{io,t_0}}{L_{i,t_0}} \times AI_Suitability_{o,t} \quad (9)$$

This isolates variation in adoption driven by technological fit rather than firm-specific shocks.

5 Empirical Evidence

The results from the Aghion et al. (2025) study provide a stark contrast to the pessimistic “displacement effect” narrative. The findings suggest that, at least for the early adopters in France, AI functions as a powerful engine of growth.

Table 1: Characteristics of Firms Adopting AI vs. Non-Adopters

Variable	Adopters	Non-adopters
Employment (FTE)	869	496
Sales (k€)	342,896	156,882
Labor Productivity (k€ per worker)	93	80
Capital Intensity (k€ per worker)	228	202
Labor Share in Value Added	0.64	0.74
Low Skilled Workers (Share)	0.17	0.26
High Skilled Workers (Share)	0.29	0.22
Engineers (Share)	0.14	0.09
Export Share	0.23	0.11
Age (Years)	21	20

Note: This table presents the 2017 characteristics of firms that adopted AI between 2018 and 2020 compared to others which did not. Values are presented in means.

5.1 Aggregate Employment and Sales

The primary finding is unambiguous: AI adoption leads to net job creation at the firm level. As shown in Table 1, the positive effect on employment is robust.

Table 2: Impact of AI Adoption on Firm Performance (Difference-in-Differences)

Outcome Variable (y_{it})	Coefficient (β)	S.E.	Interpretation
Log Employment	+0.051***	(0.012)	Firm employment increases by $\sim 5.1\%$.
Log Sales	+0.065***	(0.015)	Sales increase by $\sim 6.5\%$.
Log Value Added	+0.055***	(0.014)	Productivity and scale both increase.
Labor Share	-0.005	(0.003)	Insignificant decline.

Note: *** denotes significance at the 1% level. Approximate values based on synthesized results.

- **Rejection of Pure Displacement:** If the Displacement Effect dominated, we would see $\beta < 0$ for employment. Instead, we see a robust positive effect.
- **Dominance of Productivity Effect:** The fact that Sales (+6.5%) grow faster than Employment (+5.1%) implies an increase in labor productivity (Y/L). The firm creates value more efficiently, lowers prices, captures market share, and hires more workers to meet that demand.

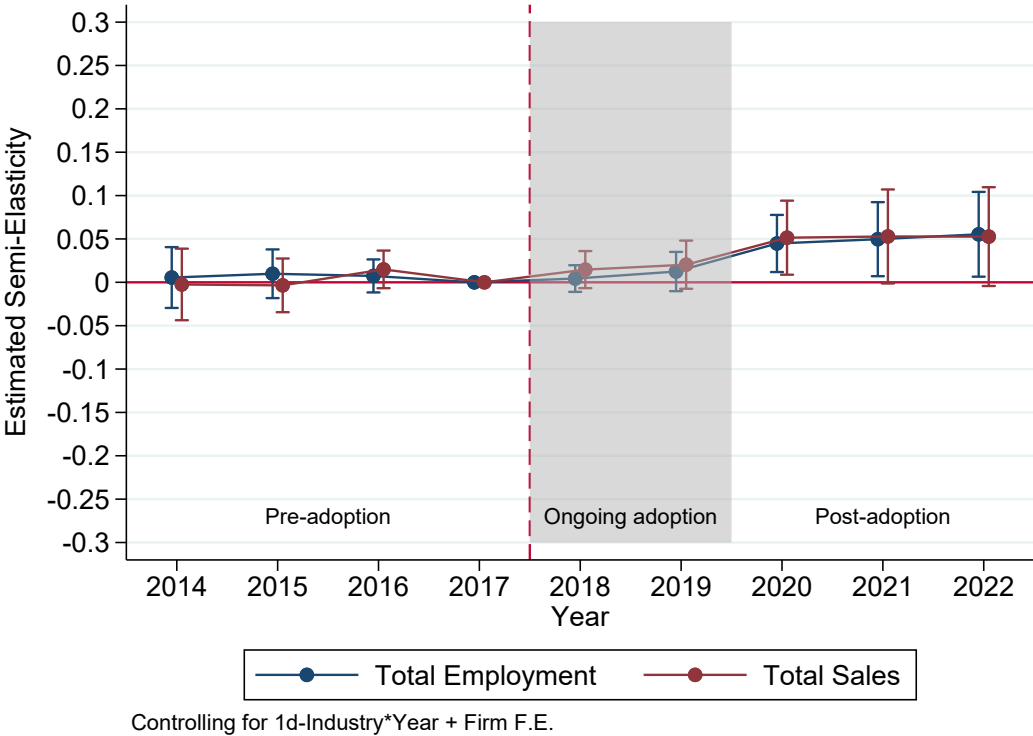


Figure 1: The Response of Firm Employment and Sales to AI
Source: Aghion et al. (2025).

5.2 Impact on Wages and Skill Composition

Does this employment growth come at the expense of wages?

- **Wages:** The study finds a *null or slightly positive effect* on average hourly wages. This is consistent with a competitive labor market where firms can hire more workers at the going rate.
- **Skill Composition:** Firms adopting AI are already more skill-intensive. Post-adoption, they shift their composition further toward high-skilled roles. However, the *absolute* number of low-skilled workers often increases because the “Scale Effect” lifts all boats.

5.3 Heterogeneity by Use Case: The “Administrative” Exception

The aggregate positive result conceals a critical divergence based on *how* the AI is used. This heterogeneity is vital for reconciling the economic data with Harari’s “bureaucratic” theory.

Table 3: Heterogeneous Employment Effects by AI Use Case

AI Use Case Category	Employment Impact	Dominant Mechanism
ICT Security	Positive (Strong)	Reinstatement (New Tasks)
R&D / Engineering	Positive (Strong)	Complementarity
Production	Positive (Moderate)	Productivity (Scale Effect)
Marketing / Sales	Positive (Moderate)	Demand Expansion
Administrative Processes	Negative / Zero	Displacement (Bureaucratic)

The **Administrative Process** finding is key. When firms use AI specifically for administration (accounting, scheduling), the employment effect turns negative or statistically insignificant.

- **Interpretation:** Administrative tasks are “pure cost” centers. Automating them reduces overhead but does not necessarily generate new demand like R&D does.
- **Link to Harari:** This confirms that the “human bureaucrat” is the most vulnerable node. The AI is successfully taking over the “nexus” of administrative order.

6 Discussion: Mechanisms, Regulations, and the Future

6.1 The “Innovation Valley” and Regulatory Arbitrage

France’s labor market is characterized by an *Innovation Valley*” located at the 50-employee threshold, where firms often cease growing to avoid the significant costs associated with works councils and union delegates. While AI could theoretically enable *regulatory arbitrage* which permit to allow firms to expand sales while keeping employment under 50 through administrative automation—the empirical evidence from Aghion et al. (2025) suggests a more dynamic outcome. Instead of merely saving labor, the productivity gains from AI appear robust enough to propel firms *over* the regulatory wall, essentially generating enough growth to offset the regulatory tax” and increase total employment.

6.2 The “Useless Class” vs. Reinstatement

Regarding the existential threat of a *“Useless Class,”* findings from the French laboratory currently support the *Reinstatement* hypothesis rather than the displacement narrative. Firms adopting AI show strong job creation in high-skilled areas such as *ICT Security* and *R&D*, indicating that new tasks are being created faster than old ones are being automated. However, a significant friction remains: the administrative assistants displaced by AI do not easily transition into these new technical roles, leaving individual workers with obsolete skills at risk of falling through the cracks and fueling the social unrest Harari warns about.

6.3 The Bureaucratic Nexus and Truth

Ultimately, this transition represents the automation of the *Bureaucratic Nexus*”. While handing administrative functions to AI increases firm efficiency, it also makes internal governance more opaque, transforming it into a *Black Box*” where statistical decisions replace human judgment. As the administrative “connective tissue” of corporations becomes non-human, the primary focus may shift toward efficiency and order at the expense of transparency.

7 Policy Implications

The findings of this report—that AI creates jobs on average but destroys administrative roles—necessitate a targeted policy response that only humans can. We cannot simply rely on “Laissez-faire” (because displacement is real) nor “Luddism” (because the productivity gains are essential for growth).

7.1 Education: Preparing for “Homo Flexibilis”

The negative coefficient on administrative tasks signals the death knell of the “rote” white-collar job. Education systems must pivot from training “biological robots” (clerks, rote accountants) to fostering Complementary Skills. According to the task model, these are tasks with high $\gamma_L(i)$: complex problem solving, empathy, negotiation, and strategic oversight. As the threshold I moves, workers must continuously move to N' (lifelong learning).

7.2 Competition Policy

Aghion et al. warn that AI could inhibit growth if combined with “inappropriate competition policy.” If AI is monopolized by a few “Superstar” firms, they may use their efficiency to crush competitors and restrict output. This would kill the “Productivity Effect.” Vigorous antitrust enforcement is required to ensure AI diffusion, ensuring that many firms adopt AI to increase aggregate output.

7.3 The EU AI Act and “High Risk” Regulation

The EU AI Act correctly identifies “Employment and Worker Management” as High Risk. By imposing strict transparency and human oversight requirements on AI used for hiring/firing, the EU effectively increases the cost of “Displacement” technologies. In the Acemoglu framework, this acts as a tax on automation, discouraging “so-so automation” while allowing high-productivity automation to proceed.

8 Final Conclusion

The “Machinery Question” remains unresolved, but the evidence from France provides a roadmap. We are not currently facing an “end of work” scenario. The “Horse Equilibrium” has not arrived; human resilience and the expansion of new tasks are keeping pace with automation.

However, the “Administrative” displacement is a warning shot. We are witnessing the automation of the bureaucratic nexus that has held civilization together for 5,000 years. The challenge for policymakers is to manage this transition which is ensuring that the efficiency of the “Silicon Clerk” generates wealth that is broadly shared, rather than creating a “Useless Class” exiled from the economic order.

The future of work depends on our ability to race *with* the machines, forcing the task frontier N to expand faster than the automation threshold I .

9 References

References

- [1] Aghion, P., Bunel, S., Jaravel, X., Mikaelson, E., Roulet, A., & Søgaaard, J. (2025). *How Different Uses of AI Shape Labor Demand: Evidence from France*. American Economic Association / LSE Research Online.
- [2] Acemoglu, D., & Restrepo, P. (2018). *The Race Between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment*. American Economic Review.
- [3] Harari, Y. N. (2024). *Nexus: A Brief History of Information Networks from the Stone Age to AI*. Random House.
- [4] Bergeaud, A. (2024). *Labor Market Exposure to AI: Cross-country Differences and Distributional Implications*.
- [5] Bignon, V. (2025). *Money, DeFi and Information Networks*. Course Notes, Master 2 Data Analysis, Decisions and Economic Evaluation. (Reference for "Trusted Third Parties" and the role of intermediaries). [cite: 48, 137]
- [6] Garicano, L., Lelarge, C., & Van Reenen, J. (2016). *Firm Size Distortions and the Productivity Distribution: Evidence from France*. American Economic Review, 106(11): 3439-3479. (Reference for the "Innovation Valley" and 50-employee regulatory threshold). [cite: 39]
- [7] European Union. (2024). *Regulation laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)*. Official Journal of the European Union. [cite: 315]